# Deep Interpretable Fully CNN Structure for Sparse Hyperspectral Unmixing via Model-Driven and Data-Driven Integration

Fanqiang Kong, Mengyue Chen, Yunsong Li, *Member, IEEE*, Dan Li, and Yuhan Zheng

***Abstract*— Hyperspectral unmixing (HSU), which aims to identify constituent materials and estimate the corresponding proportions in a scene, is an essential research topic in remote sensing. Most deep learning-based methods are data-inspired, relying on massive amounts of data to train black-box-like networks, while a few model-inspired unmixing networks only consider the spectral features of the pixel, ignoring the exploration of spatial information between pixels. In this article, we design a network topology according to the classical iterative algorithm, and the large number of learnable parameters contained in the network is continuously updated through data fitting. In other words, we integrate the concepts of both model- and data-driven and propose a deep interpretable fully convolutional neural network (DIFCNN). The iteration of the classic sparse unmixing algorithm is unfolded to provide guidance for the network structure and incorporate prior knowledge into the network. Meanwhile, 2-D convolutional layers are employed to automatically learn the spatial information at different scales. A known spectral library is used as a prior to initialize network parameters and reconstruct the image. The DIFCNN adopts an end-to-end training strategy; in addition, we establish a new loss function that adds a joint sparse constraint on the abundance result to the cross-entropy loss. Experiments on both synthetic and real datasets show that the performance of the DIFCNN not only outperforms the sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) and its improved algorithms but also is highly competitive in the state-of-the-art methods of deep learning.**

***Index Terms*— Algorithm unrolling, convolutional neural network (CNN), hyperspectral unmixing (HSU), model-driven, spectral–spatial.**

## I. Introduction

With the advantage of containing abundant spectral information, HSIs have been applied in various fields, such as military reconnaissance, environmental monitoring, mineral exploration, and soil investigation [1] citeEnhanced. However, higher spectral resolution means lower spatial resolution, which is determined by the inherent limitations of spaceborne hyperspectral sensors. There are often multiple ground objects in the area corresponding to a pixel, so its spectral curve is a mixture of the reflection spectra of these objects, and such a pixel is also called "mixed pixel." The existence of mixed pixels inevitably hinders the accuracy of image postprocessing analysis and is unfavorable to the quantitative development of remote sensing. To address this issue, hyperspectral unmixing (HSU) serves as an effective solution that aims to separate the mixed spectral curve into its pure components (endmembers) and their corresponding proportions (abundances) in the region.

According to the mixing modes of endmember signatures, there are two mixture models on which all the unmixing methods are based, namely, linear spectral mixture model (LMM) and nonlinear spectral mixture model (NLMM) [2]. LMM is based on the assumption that the reflectance spectrum of each pixel in the HSI is a linearly weighted combination of the associated endmember signatures, while NLMM considers nonlinear situations and is dedicated to describing natural real scenes. LMM is currently the most widely used model in the field of unmixing because of its simplicity, physical interpretability, and easy calculation.

Traditional LMM-based methods are divided into three categories, geometrical-based, statistical-based, and sparsity-based [3]. Among them, the geometrical-based approaches are subdivided into pure pixel-based algorithms [4], [5] and minimum volume-based algorithms [6], while neither of them can cope with the situation when the spectra of materials are highly mixed, because of the insufficiency of spectral vectors included in the simplex facets. At this point, the statistical-based method is a good alternative, at the cost of higher computational complexity. Its representative algorithms are nonnegative matrix factorization (NMF) [7] and its variants [8], [9]. NMF can estimate endmembers and abundances simultaneously without the pure pixel hypothesis but suffers from the problem of extracting virtual endmembers that have no physical meaning. There is no such concern in the sparsity-based methods. Sparse unmixing represents the observed HSI as a linear combination of many endmember signatures from a known spectral library, that is, endmembers are no need to be extracted from the HSI, and endmember contained in the library has clear physical meaning. Typically, sparsity constraints related to abundance are added to the optimization problem and solved using the alternating direction method of multipliers (ADMM). For instance, $l_1$ regularization is

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imposed on the abundance in the sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) [10] and $l_{2,1}$ regularization for collaborative SUnSAL (CLSUnSAL) [11]. Considering the existence of spatial contextual information in HSI, the related assumption of spatial correlation is added to the constraints as prior information. Total variation (TV) regularization in the SUnSAL-TV [12] assumes that adjacent pixels have similar abundance results, which enforces the spatial uniformity between adjacent pixels; a spectral–spatial weighted sparse unmixing ($S^2$WSU) [13] framework uses both spectral and spatial weighting factors, providing flexibility for the exploration of different spatial scenarios. Considering the redundancy of the spectral library, greedy algorithms, such as subspace matching pursuit (SMP) [14], extract effective endmember subset from the spectral library through matching pursuit. However, the performance is greatly dependent on the accurate extraction of the subset, which is extremely challenging in a spectral library with a large number of similar endmembers.

Deep learning has advanced quickly during the decade in various areas of remote sensing image processing, including band selection [15], [16], spectral images fusion [17], [18], [19], and HSU [20]. Most current neural networks used in unmixing tasks are designed based on the data-driven, which is popular for its ease of design. Data-driven networks are classified into three types, supervised, unsupervised, and semisupervised, depending on the way they are trained. The supervised approaches [21], [22] require true abundance in advance as labels during training. However, due to the difficulty in obtaining real abundance data, the supervised approaches are rarely studied nowadays. The unsupervised networks are usually encoder–decoder structures where the encoder maps the input into abundance and the decoder outputs the reconstruction of HSI. Several autoencoder (AE) networks proposed in earlier years are designed only for the deep interspectral characteristics of hyperspectral images, including stacked AE [23], variational AE [24], and denoising AE [25]. To better exploit the spectral information of HSI, some methods impose constraints on the parameters of the network or construct specific loss function [26], and some methods design specific structures, such as Deep HSNet [27] and SCA-Net [28]. Taking the image patch or whole image as input instead of the spectral vector of a pixel can improve the accuracy of unmixing by extracting the spatial information in the HSI. MTAEU [29] utilizes multitask learning to obtain spatial correlation among adjacent pixels. Convolutional neural network (CNN)-based networks are widely applied due to their ability to learn spatial information, e.g., CNNAEU [30] and CyCU-Net [31]. There are also methods for feature extraction in different dimensions of the image. Multibranch CNNs employ 1-D, 2-D, and 3-D convolutions simultaneously to extract features from the spectral, spatial, and spectral–spatial dimensions of the HSI patch [32]; GDBAN [33] uses gated dual-branch CNN to exploit local spatial context and spectral characteristic of central pixel. In addition to integrating spatial information, researchers have worked on exploiting other information to improve unmixing accuracy. For example, MiSiCNet [34] incorporates geometrical information, while UnDIP [35] introduces the deep image prior to transform input noise into abundance maps. The accurate estimation of endmembers is a major obstacle for unsupervised approaches. Some methods utilize additional networks to provide guidance, such as EGU-Net [36], whose endmember-guided network shares the weights in the representation of endmembers with the main unmixing network; GAUSS [37] uses one-hot encoded abundances as the pseudo-ground truth to guide the unmixing network. Other types of networks, such as long short-term memory (LSTM) network [38], transformers [39], and generative networks [40], all have applications in the field of unsupervised unmixing. For the semisupervised networks without endmember extraction, pixel/cube-based DCAE [41] extracts endmembers through vertex component analysis (VCA) and the network is only used to estimate abundance. The accessible spectral library is also commonly adopted in semisupervised unmixing. A deep spectral convolution network incorporating spectral library called SCNL [42] is allowed to be used for a series of HSIs; SUnCNN [43] realizes the first application of deep learning on sparse unmixing using 2-D convolutional layers and skip connection. These data-driven networks rely on large amounts of data to train and the structure of the network lacks interpretability.

On the contrary, model-driven networks often have careful theoretical analysis, mathematical derivation, and strong interpretability. When the established model is accurate enough, the quality of the solution will far exceed expectations. However, it has the limitation of difficulty in building accurate models due to unpredictable factors in practical applications. At present, model-driven networks are rarely used in HSU. Similar to data-driven, model-driven approaches are mainly applied in unsupervised unmixing tasks. For instance, MNN-BU [44] is the first attempt and it unfolds the iteration of the iterative shrinkage-thresholding algorithm (ISTA) into an interpretable layer. Another example is the group sparsity regularized unmixing unrolling (GSUU) network [45], which unrolls a regularized matrix factorization objective function for blind unmixing. On the other hand, several works have conducted experiments on both unsupervised and semisupervised unmixing tasks, such as SNMF-Net [46] and ADMM-Net [47]. The former makes the endmembers no longer represented by the weights of the decoder, but the output at the same time as the abundance results, and the latter unrolls the ADMM model, which is a new attempt of model-driven approach in the HSU field. IVIU-Net [48] focusing on semisupervised unmixing has also been studied recently. To the best of our knowledge, in order to intuitively represent the specific operations of the algorithm, the current deep unrolling-based unmixing methods are pixelwise, the spatial information in the HSI is ignored, and the fully connected (FC) layers are employed to represent the multiplication operation.

Model-driven can compensate for the lack of interpretability in data-driven, and the variability of structures in data-driven gives the probability to extract the spatial–spectral joint features, which is ignored in model-driven. Moreover, deep learning-based methods have made little progress in sparse

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unmixing compared to blind unmixing. In this article, we integrate the concepts of both model-driven and data-driven and propose a deep interpretable fully CNN (DIFCNN) structure for sparse unmixing. The contributions can be summarized as follows.

1) It is the first attempt to combine the unrolling network with spatial information extraction. A new means of convolutional layer initialization is proposed to obtain an initial state consistent with the multiplication operation, which ensures that spatial information can be extracted without affecting interpretability.
2) Unlike existing model-driven networks where the settings of each layer are the same, the encoder of DIFCNN consists of three layers in series with different convolutional kernel sizes, enabling multiscale feature extraction. A nonlinear activation function is introduced after the convolutional layers, which can speed up the convergence of the network and enhance the expressive ability of the network.
3) Considering the specificity of sparse unmixing, we construct a loss function consisting of cross-entropy loss and sparsity regularization. The cross-entropy loss is utilized to measure the discrepancy between inputs and outputs. The $l_{2,1}$ regularization on the abundance further constrains the results to structured sparsity.

The remaining part of this article proceeds as follows. Section II begins by laying out the theoretical dimensions of the research and looks at the study of algorithm unrolling technology. Section III is concerned with the explanation of the innovative initialization method and the proposed interpretable network. In Section IV, we analyze the results for both synthetic and real HSIs and compare with other state-of-the-art methods. Finally, we conclude this article and envision future research directions in Section V.

## II. Related Works

In this section, we introduce the basic concepts of sparse unmixing and related techniques used in the proposed network.

### A. Linear Spectral Mixture Model

Generally, given an HSI with $L$ wavelength-indexed bands, the LMM can be expressed as

$$\boldsymbol{y}_i = \sum_{j=1}^{p} x_{i,j}\boldsymbol{m}_j + \boldsymbol{n}_i = \boldsymbol{M}\boldsymbol{x}_i + \boldsymbol{n}_i \tag{1}$$

where $\boldsymbol{y}_i \in \mathbb{R}^{L\times 1}$ is the $i$th observed mixed pixel, $\boldsymbol{m}_j \in \mathbb{R}^{L\times 1}$ denotes the spectrum of the $j$th endmember, $\boldsymbol{x}_i \in \mathbb{R}^{p\times 1}$ and $\boldsymbol{n}_i \in \mathbb{R}^{L\times 1}$ are the vectors of fractional abundance and additive noise, respectively, corresponding to the pixel, and $\boldsymbol{M} = [\boldsymbol{m}_1, \boldsymbol{m}_2 \ldots, \boldsymbol{m}_p] \in \mathbb{R}^{L\times p}$ is the matrix of $p$ endmembers. For the whole HSI $\boldsymbol{Y} \in \mathbb{R}^{L\times N}$ with $N$ pixels, the matrix notation of (1) is

$$\boldsymbol{Y} = \boldsymbol{M}\boldsymbol{X} + \boldsymbol{N} \tag{2}$$

in which $\boldsymbol{X} \in \mathbb{R}^{p\times N}$ and $\boldsymbol{N} \in \mathbb{R}^{L\times N}$ are separately abundance matrix and noise matrix, respectively.

Since the fractional abundance vector for each pixel signifies the occupancy of all involved endmembers, in order to make the unmixing results physically meaningful, two constraints are always imposed on the abundance matrix, namely, abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC). Their specific mathematical representation is given as follows:

$$\boldsymbol{X} \geq \boldsymbol{0}, \quad \boldsymbol{1}_p^T\boldsymbol{X} = \boldsymbol{1}_N^T \tag{3}$$

where $\boldsymbol{1}$ is the all-ones vector. In addition, the endmember matrix is also constrained to be nonnegative.

### B. Sparse Unmixing

When the endmember matrix is replaced by a known spectral library $\boldsymbol{A} \in \mathbb{R}^{L\times m}$, that is, a large number of irrelevant spectra are introduced into the endmembers, the solution of the abundance matrix can be regarded as a sparse regression problem. Typically, the number of endmembers in the spectral library is much larger than the number of true endmembers in the HSI, i.e., $m \gg p$, and thence, the abundance matrix needs to be sparse. The sparse unmixing problem can be expressed as the following $l_2 - l_0$ optimization problem:

$$\min_{\boldsymbol{X}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{A}\boldsymbol{X}\|_F^2 + \lambda\|\boldsymbol{X}\|_0, \quad \text{s.t. } \boldsymbol{X} \geq 0 \tag{4}$$

where $\|\cdot\|_F$ represents the Frobenius norm of the matrix, $\lambda$ is the regularization parameter, and $\|\cdot\|_0$ is the $l_0$-norm. The $l_0$-norm denotes the number of nonzero atoms in the matrix, which is discrete and nonconvex, so the formula (4) is a typical NP-hard problem. In this case, the optimization problem cannot be solved in a straightforward way, and the greedy algorithms are usually utilized for endmember extraction.

The convex optimization method can be used to solve the problem after replacing the abundance $l_0$-norm constraint in (4) with a convex sparse constraint

$$\min_{\boldsymbol{X}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{A}\boldsymbol{X}\|_F^2 + \lambda\|\boldsymbol{X}\|_{p,1}, \quad \text{s.t. } \boldsymbol{X} \geq 0. \tag{5}$$

When $p = 1$, $\|\boldsymbol{X}\|_{1,1} = \sum_{j=1}^{N}\|\boldsymbol{x}_j\|_1$, it becomes a constrained sparse regression (CSR) problem, which can be solved efficiently by solving several separate subproblems using ADMM. This method is the classic SUnSAL algorithm. Under the single measurement vector (SMV) model, the equivalent form of the optimization function is defined as

$$\min \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|_F^2 + \lambda\|\boldsymbol{x}\|_1 + \iota_{\mathbb{R}^+}(\boldsymbol{x}) \tag{6}$$

where

$$\iota_{\mathbb{R}^+}(\boldsymbol{x}) = \begin{cases} 0, & \boldsymbol{x} \geq 0 \\ \infty, & \boldsymbol{x} < 0. \end{cases}$$

Then, split (6) as two subproblems, $f_1(\boldsymbol{x}) = (1/2)\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|_F^2$ and $f_2(\boldsymbol{u}) = \lambda\|\boldsymbol{u}\|_1 + \iota_{\mathbb{R}^+}(\boldsymbol{u})$, where $\boldsymbol{x} = \boldsymbol{u}$. The $(k+1)$th iterative steps through ADMM are

$$\boldsymbol{x}_{k+1} = (\boldsymbol{A}^T\boldsymbol{A} + \mu\boldsymbol{I})^{-1}(\boldsymbol{A}^T\boldsymbol{y} + \mu(\boldsymbol{u}_k + \boldsymbol{d}_k)) \tag{7}$$

$$\boldsymbol{u}_{k+1} = \max\left\{0, \text{soft}\left((\boldsymbol{x}_{k+1} - \boldsymbol{d}_k), \frac{\lambda}{\mu}\right)\right\} \tag{8}$$

$$\boldsymbol{d}_{k+1} = \boldsymbol{d}_k - (\boldsymbol{x}_{k+1} - \boldsymbol{u}_{k+1}) \tag{9}$$

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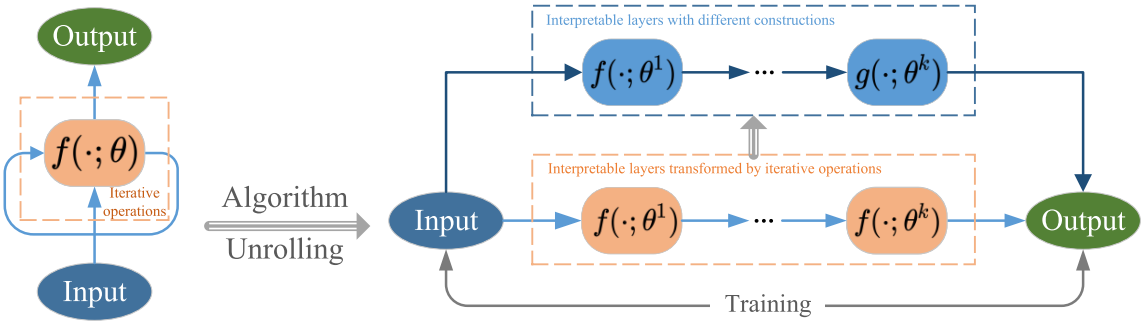



Fig. 1. Schematic of the algorithm unrolling structure.

where $\mu \geq 0$, $\boldsymbol{d}$ is the dual variable, and soft(, ) is the soft-threshold operator.

When $p = 2$, $\|\boldsymbol{X}\|_{2,1} = \sum_{j=1}^{N} \|\boldsymbol{x}_j\|_2$, it becomes CLSUnSAL. The $l_{2,1}$-norm imposes a collaborative sparsity constraint on all pixels simultaneously, which is based on the prior that only a few endmembers of the spectral library exist in real HSIs. Compared with SUnSAL, CLSUnSAL utilizes collaborative sparsity of the image.

In order to further exploit the spatial contextual information in HSIs, TV regularization is introduced to the optimization function. It assumes that spatially adjacent pixels have similar abundances. The well-known SUnSAL-TV algorithm incorporates the SUnSAL and TV regularization, which is represented as

$$\min_{\boldsymbol{X}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{A}\boldsymbol{X}\|_F^2 + \lambda\|\boldsymbol{X}\|_{1,1} + \lambda_{\mathrm{TV}}\mathrm{TV}(\boldsymbol{X}) \quad \text{s.t } \boldsymbol{X} \geq 0 \tag{10}$$

in which $\mathrm{TV}(\boldsymbol{X}) = \sum_{\{i,j\}\in\varepsilon} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_1$, where $\varepsilon$ is the set of horizontal and vertical neighbors in the image. The TV regularization makes the abundance difference between pixels in the neighborhood smaller, and in other words, it smoothies the abundance maps.

### C. Algorithm Unrolling Technique

It not only retains the advantages of the model-driven approach [49] but also utilizes the powerful data fitting ability of deep learning. It designs the network topology according to the iterative algorithm for solving the rough model established by the problem and tunes the weights of the network through data fitting.

Fig. 1 shows the general process of algorithm unrolling, where $\theta$ represents the parameters generated by objective function or prior knowledge. In the iterative algorithm, $\theta$ is fixed. After unrolling it to the deep network, these parameters of each layer $\theta^i$ are set as learnable parameters initialized by $\theta$, and the training process is to learn the appropriate $\theta^i$ from the data. To sum up, the topology of the network depends on the algorithm derived from the model, while the parameters are learned from the data through deep learning. While a normal algorithm unrolling network is a stack of identical iterative layers, the network we designed uses different structures in different layers to improve flexibility, which proves to be effective in Section IV.

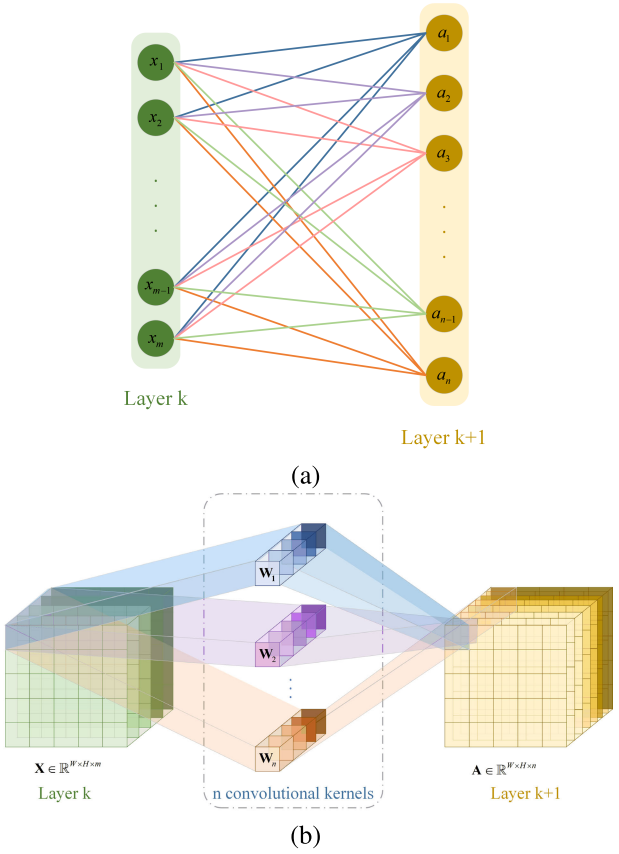



Fig. 2. Diagrams of (a) Linear FC layer and (b) $1 \times 1$ convolutional layer.

## III. Deep Interpretable CNN Structure

In this section, we demonstrate a fully CNN structure inspired by the SUnSAL algorithm. Besides, the proposed unrolling layer is explained in detail, as well as the particular initialization strategy.

### A. Incorporating Spatial Information

In order to effectively extract the spatial information of the image during the training process, the input of the proposed network is the 3-D HSI patch instead of the pixel vectors. In addition, we also improve the way of parameters initialization and expand FC layers to $n \times n$ convolutional layers to obtain a larger receptive field.

First, the matrix multiplication operation can be replaced by the linear layer without bias in the network. Fig. 2(a) shows

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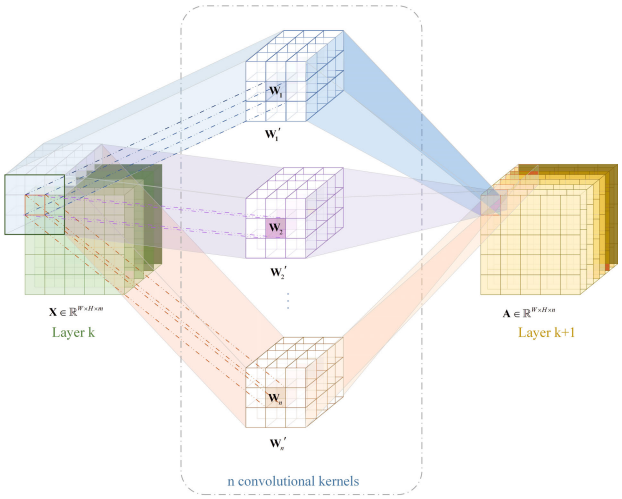



Fig. 3. Operation of convolution.

the process of one FC layer. Suppose that for the $k$th FC layer, the input is $\boldsymbol{x} \in \mathbb{R}^{m\times 1}$, and the output is $\boldsymbol{a} \in \mathbb{R}^{n\times 1}$; then, the operation can be represented by the matrix form

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{n-1} \\ a_n \end{bmatrix} = \begin{bmatrix} \omega_{11} & \omega_{12} & \cdots & \omega_{1m} \\ \omega_{21} & \omega_{22} & \cdots & \omega_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{(n-1)1} & \omega_{(n-1)2} & \cdots & \omega_{(n-1)m} \\ \omega_{n1} & \omega_{n2} & \cdots & \omega_{nm} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \tag{11}$$

where $\boldsymbol{W} \in \mathbb{R}^{n\times m}$ is the weight of this layer.

When the input is a 3-D patch $\boldsymbol{X} \in \mathbb{R}^{W\times H\times m}$, $H$ and $W$ are the height and width of the spatial domain, respectively, and $m$ is the channel size. In order to do the same operation as the FC layer on each pixel and make the size of output feature maps become $W \times H \times n$, a $1 \times 1$ convolutional layer should be introduced. The related process is shown in Fig. 2(b). The $i$th convolution kernel $\boldsymbol{W}_i = [[\omega_{i1}, \omega_{i2}, \ldots, \omega_{im}]] \in \mathbb{R}^{m\times 1\times 1}$ corresponds to the $i$th row in the FC layer weight. Therefore, the value of the neurons in the $(k+1)$th layer and the $i$th feature map is

$$\boldsymbol{a}_i^{yz} = g\left(\sum_j \omega_{ij} x_j^{yz} + b_i\right) \tag{12}$$

where $(y, z)$ is the spatial position, $j$ indexes the feature map in the $k$th layer, $b_i$ is the bias, and $g(\cdot)$ is the activation function. If the bias and activation function are not used in the network, the $1 \times 1$ convolutional layer is equivalent to an FC layer acting on each pixel.

Next, extend $1 \times 1$ convolution to $n \times n$ convolution. Taking $3 \times 3$ convolution as the example, the process of $3 \times 3$ convolution is shown in Fig. 3. To achieve the same effect as the above two operations in the initial state, both padding and stride are set to one, no bias and activation function are used, and the filters become

$$\boldsymbol{W}_i' = \begin{bmatrix} \boldsymbol{\omega}_i'^{(-1)(-1)} & \boldsymbol{\omega}_i'^{(-1)0} & \boldsymbol{\omega}_i'^{(-1)1} \\ \boldsymbol{\omega}_i'^{0(-1)} & \boldsymbol{\omega}_i'^{00} & \boldsymbol{\omega}_i'^{01} \\ \boldsymbol{\omega}_i'^{1(-1)} & \boldsymbol{\omega}_i'^{10} & \boldsymbol{\omega}_i'^{11} \end{bmatrix}.$$

At this time, the value of neurons in the $(k+1)$th layer and the $i$th feature map is

$$\begin{aligned} \boldsymbol{a}_i^{yz} &= g\left(\sum_j \sum_{m=-\frac{M_i-1}{2}}^{\frac{M_i-1}{2}} \sum_{n=-\frac{N_i-1}{2}}^{\frac{N_i-1}{2}} \omega_{ij}'^{mn} x_j^{(y+m)(z+n)} + b_i\right) \\ &= g\left(\sum_j \sum_{m=-1}^{1} \sum_{n=-1}^{1} \omega_{ij}'^{mn} x_j^{(y+m)(z+n)} + b_i\right) \end{aligned} \tag{13}$$

in which $M_i$ and $N_i$ are the length and width of the kernel size, respectively, $\boldsymbol{W}' = [\boldsymbol{W}_1', \boldsymbol{W}_2', \ldots, \boldsymbol{W}_n'] \in \mathbb{R}^{n\times m\times 3\times 3}$ is the new weight, $\boldsymbol{W}_i' \in \mathbb{R}^{m\times 3\times 3}$, and $\boldsymbol{\omega}'^{ab} = [\omega_1'^{ab}, \omega_2'^{ab,\ldots,\omega_m'^{ab}}] \in \mathbb{R}^{m\times 1}$.

Equation (13) is then converted to

$$\begin{aligned} \boldsymbol{a}_i^{yz} &= \sum_j \sum_{m=-1}^{1} \sum_{n=-1}^{1} \omega_{ij}'^{mn} x_j^{(y+m)(z+n)} \\ &= \sum_j \omega_{ij}'^{00} x_j^{yz} + \sum_j \sum_{\substack{m=-1 \\ (m,n)\neq(0,0)}}^{1} \sum_{n=-1}^{1} \omega_{ij}'^{mn} x_j^{(y+m)(z+n)}. \end{aligned} \tag{14}$$

Combining (12) and (14), in the initial stage, when $\boldsymbol{\omega}'^{00} = \boldsymbol{W}_i$ and $\boldsymbol{\omega}'^{ab}{}_{(a,b)\neq(0,0)} = \boldsymbol{0}$, these two formulas are equivalent. The difference is that after training the network, the $1 \times 1$ filters are only adjusted according to a single pixel, while $\boldsymbol{\omega}'^{ab}{}_{(a,b)\neq(0,0)}$ in $3 \times 3$ filters is no longer zero, which contain both spectral and spatial information of the center and its adjacent pixels.

### B. Unrolling the Iterative Step to a Network Layer

Since the SUnSAL algorithm utilizes three steps for alternating iterations, the unrolling layer also contains three sections according to (7)–(9). The complete $k$th algorithm unrolling layer is shown in Fig. 4. The dark blue lines, green lines, and orange lines indicate the update process of $\boldsymbol{X}$, $\boldsymbol{U}$, and $\boldsymbol{D}$, respectively.

The update process of $\boldsymbol{X}^k$ is derived from (7) with some improvements. First, from the perspective of 3-D image rather than 1-D pixel vector, the vector $\boldsymbol{x}$, $\boldsymbol{u}$ and $\boldsymbol{d}$ are substituted for $\boldsymbol{X}, \boldsymbol{U}, \boldsymbol{D} \in \mathbb{R}^{W\times H\times m}$. In this case, $1 \times 1$ convolution should be employed to represent multiplication, and (7) is rewritten as

$$\boldsymbol{X}^k = \boldsymbol{W} \otimes \boldsymbol{Y} + \boldsymbol{B} \otimes \left(\boldsymbol{U}^{k-1} + \boldsymbol{D}^{k-1}\right) \tag{15}$$

where $\otimes$ denotes the convolution operation, and the kernel weights $\boldsymbol{W}$ and $\boldsymbol{B}$ are calculated from the known spectral library $\boldsymbol{A}$

$$\begin{aligned} \boldsymbol{W} &= \left(\boldsymbol{A}^T\boldsymbol{A} + \mu\boldsymbol{I}\right)^{-1}\boldsymbol{A}^T \\ \boldsymbol{B} &= \left(\boldsymbol{A}^T\boldsymbol{A} + \mu\boldsymbol{I}\right)^{-1}\mu. \end{aligned} \tag{16}$$

It should be noted that $\boldsymbol{W}$ and $\boldsymbol{B}$ are to be reshaped into 4-D tensors to keep the convolution kernel size consistent. Therefore, the size of $\boldsymbol{W}$ and $\boldsymbol{B}$ is $m \times L \times 1 \times 1$ and $m \times m \times 1 \times 1$, respectively.

Expand the $1 \times 1$ convolutional layer to $n \times n$ convolutional layer, and convolution kernels $\boldsymbol{W}'^k$ and $\boldsymbol{B}'^k$ are

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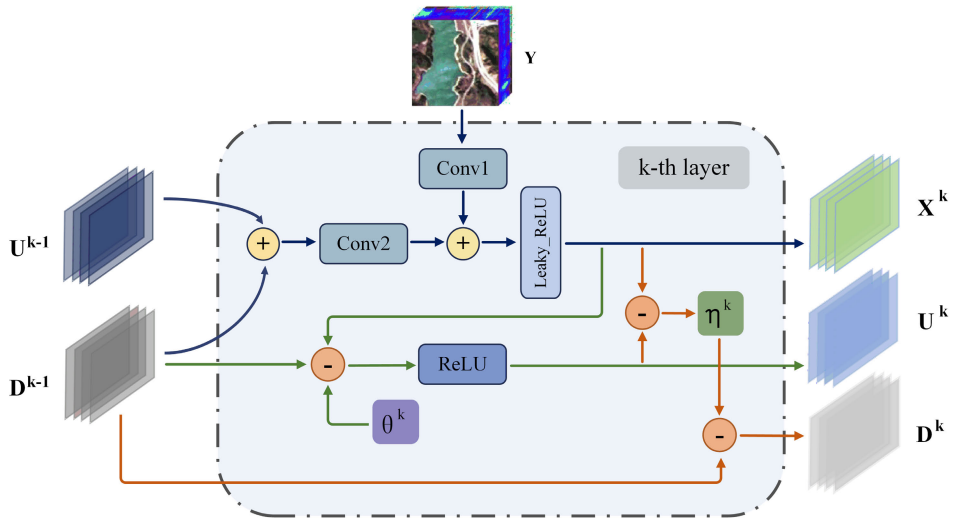



Fig. 4. Topology of the $k$th unrolling layer.

generated. Given the input of the $k$th layer $\boldsymbol{Y}$, $\boldsymbol{U}^{k-1}$, and $\boldsymbol{D}^{k-1}$, $\boldsymbol{X}^k$ can be represented as

$$\begin{aligned}\boldsymbol{X}^k &= \text{LeakyReLU}\left(\boldsymbol{W}'^k \otimes \boldsymbol{Y} + \boldsymbol{B}'^k \otimes \left(\boldsymbol{U}^{k-1} + \boldsymbol{D}^{k-1}\right)\right) \\ &= \text{Layer}_X\left(\boldsymbol{Y}, \boldsymbol{U}^{k-1}, \boldsymbol{D}^{k-1}; \boldsymbol{W}'^k, \boldsymbol{B}'^k\right)\end{aligned} \tag{17}$$

where $\boldsymbol{W}'^k$ and $\boldsymbol{B}'^k$ are learnable parameters. Meanwhile, the nonlinear leaky rectified linear unit (ReLU) activation function is additionally added after the convolution operations to accelerate the calculation and the convergence of the network. Meanwhile, the leaky ReLU suppresses negative results in $\boldsymbol{X}^k$, forming a nonnegativity constraint on the results to a certain extent.

The update of $\boldsymbol{U}^k$ is derived from (8), which can be simplified as $\boldsymbol{u}_{k+1} = \max\{0, \boldsymbol{x}_{k+1} - \boldsymbol{d}_k - (\lambda/\mu)\}$, given $\boldsymbol{X}^k$ and $\boldsymbol{D}^{k-1}$, and this part can be expressed as

$$\begin{aligned}\boldsymbol{U}^k &= \text{ReLU}\left(\boldsymbol{X}^k - \boldsymbol{D}^{k-1} - \theta^k \boldsymbol{I}\right) \\ &= \text{Layer}_U\left(\boldsymbol{X}^k, \boldsymbol{D}^{k-1}; \theta^k\right)\end{aligned} \tag{18}$$

where $\theta^k$ is a learnable parameter learned by network training to better fit the unmixing problem. The ReLU function replaces $\max(0, x)$, ensuring the nonnegativity of the output.

The third update component is $\boldsymbol{D}^k$, given the input $\boldsymbol{D}^{k-1}$, $\boldsymbol{X}^k$, and $\boldsymbol{U}^k$, and the update formula is derived from (9)

$$\begin{aligned}\boldsymbol{D}^k &= \boldsymbol{D}^{k-1} - \eta^k\left(\boldsymbol{X}^k - \boldsymbol{U}^k\right) \\ &= \text{Layer}_D\left(\boldsymbol{X}^k, \boldsymbol{U}^k, \boldsymbol{D}^{k-1}; \eta^k\right)\end{aligned} \tag{19}$$

where a learnable parameter $\eta^k$ is set before $(\boldsymbol{X}^k - \boldsymbol{U}^k)$ to obtain a weight, which is more suitable for the current unmixing task.

In summary, the $k$th unrolling layer can be comprehensively expressed as

$$\boldsymbol{X}^k, \boldsymbol{U}^k, \boldsymbol{D}^k = \text{Layer}\left(\boldsymbol{Y}, \boldsymbol{U}^{k-1}, \boldsymbol{D}^{k-1}; \boldsymbol{W}'^k, \boldsymbol{B}'^k, \theta^k, \eta^k\right) \tag{20}$$

that is, in the $k$th layer, only $\{\boldsymbol{W}'^k, \boldsymbol{B}'^k, \theta^k, \eta^k\}$ are trainable parameters, and the rest are the shortcuts of each component and skipping connection from the input $\boldsymbol{Y}$.

TABLE I
HYPERPARAMETERS USED IN DIFCNN

| | Input Ch. | Output Ch. | Filter Size | Stride |
|---|---|---|---|---|
| Conv0 | $L$ | $m$ | 3×3 | 1 |
| Conv11 | $L$ | $m$ | 5×5 | 1 |
| Conv12 | $m$ | $m$ | 5×5 | 1 |
| Conv21 | $L$ | $m$ | 3×3 | 1 |
| Conv22 | $m$ | $m$ | 3×3 | 1 |
| Conv31 | $L$ | $m$ | 1×1 | 1 |
| Conv32 | $m$ | $m$ | 1×1 | 1 |
| Conv4 | $m$ | $L$ | 1×1 | 1 |
| Optimizer | Type: Adam | | Learning Rate: 1e-4 | |

### C. Sparse Unmixing Network Structure

It is aforementioned that an unrolling layer can represent an iteration of traditional algorithm, and then, a complete algorithm iteration can be expressed by the concatenation of $n$ unrolling layers. Since the parameters in the layers are learnable, only a few layers are needed.

As shown in Fig. 5, we construct a fully CNN network, which is only composed of three iterative layers. Table I lists the related settings used in the DIFCNN, and the parameters of each layer are trained independently. The network has a stable structure because the number of feature maps output by each layer depends only on the number of $L$ spectral bands in HSIs and the number of endmembers $m$ in the spectral library. Different sizes of convolutional kernels, i.e., $5 \times 5$, $3 \times 3$, and $1 \times 1$, are separately employed in different iterating layers to realize spatial information extraction in different receptive field sizes. Structurally, it is equivalent to performing multiscale feature extraction on the input image.

After three iterating layers, the outputs $\{\boldsymbol{X}^3, \boldsymbol{U}^3, \boldsymbol{D}^3\}$ are obtained. $\boldsymbol{U}^3$ is chosen to be abundance output for further processing. Due to the ReLU function in (18), $\boldsymbol{U}^3 \geq \boldsymbol{0}$, the result satisfies the ANC constraint. The ASC constraint is also realized through the $l_1$ normalization, and for the spatial coordinates $(i, j)$, the final abundance result is

$$\hat{\boldsymbol{X}}_{i,j} = \frac{\boldsymbol{U}^3_{i,j}}{\sum_{k=0}^{m} \boldsymbol{U}^3_{i,j,k}}. \tag{21}$$

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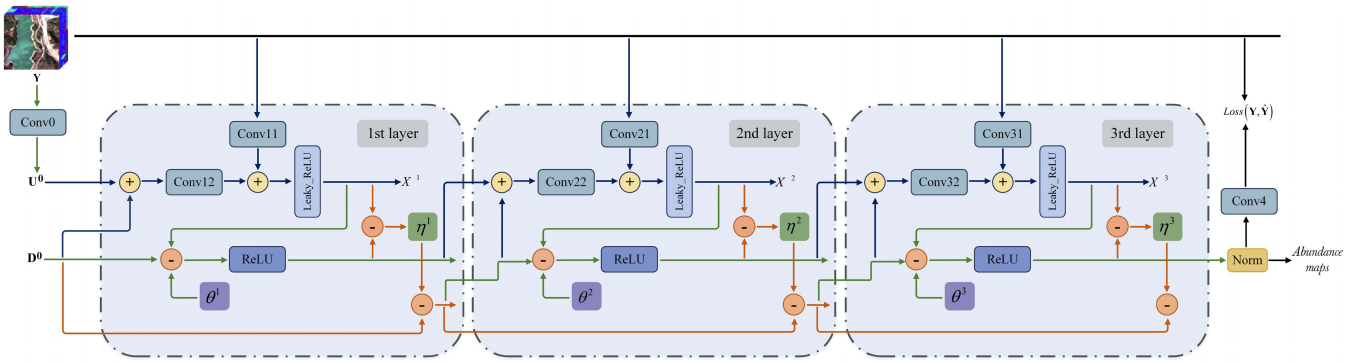



Fig. 5. Architecture of DIFCNN.

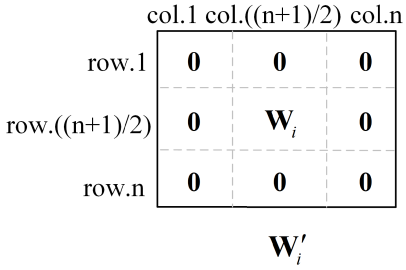



Fig. 6. Method of initializing $n \times n$ convolution kernel.

A $1 \times 1$ convolutional layer without bias is appended after the abundance output to reconstruct the image, where the kernel weight is fixed to the spectral library to meet with the LMM in (2)

$$\hat{\boldsymbol{Y}} = \boldsymbol{A} \otimes \hat{\boldsymbol{X}}. \tag{22}$$

Finally, the network parameter optimization is achieved by constructing a loss function for the input $\boldsymbol{Y}$ and the output $\hat{\boldsymbol{Y}}$ for end-to-end training.

### D. Initialization and Training Approaches

*1) Parameters Initialization:* The parameter initialization is an essential part of the DIFCNN because a proper initialization can speed up network convergence and get better results. The pseudo input of the network, $\boldsymbol{U}^0$ and $\boldsymbol{D}^0$, is derived from the SUnSAL algorithm, and based on this, it was improved by combining CNN with a $3 \times 3$ convolutional layer, $\boldsymbol{U}^0 \leftarrow \boldsymbol{W}'^0 \otimes \boldsymbol{Y}$ and $\boldsymbol{D}^0 \leftarrow 0$. $\boldsymbol{W}'^n$ and $\boldsymbol{B}'^n$ abide by the following initialization method.

First, $\boldsymbol{W}$ and $\boldsymbol{B}$ in (16) are calculated from the known spectral library $\boldsymbol{A}$ and a hyperparameter $\mu$, where the setting of $\mu$ is associated with the traditional SUnSAL algorithm. As shown in Fig. 6, expand the size of the convolution kernel to $n \times n$ by filling zeros around, and the new tensors $\boldsymbol{W}'^k \in \mathbb{R}^{m \times L \times n \times n}$ and $\boldsymbol{B}'^k \in \mathbb{R}^{m \times m \times n \times n}$ are initialized. At this time, the convolution kernels do not contain spatial information. With the training of the network, the convolution kernels automatically learn the weights of the spatial features in the $n \times n$ neighborhood, which makes the outputs include the contribution of the neighborhood features. Furthermore, $\theta^i$ is initialized to $(\lambda/\mu)$ and $\eta^i$ is initialized to 1, corresponding to the settings in (8) and (9).

*2) Network Training:* The proposed network is trained in an unsupervised way due to its encoder–decoder structure, and we use the cross-entropy function as a loss function to measure the discrepancy between the input and the output

$$\begin{aligned} L_{\text{ce}}(\boldsymbol{y}, \hat{\boldsymbol{y}}) &= -\left[\boldsymbol{y} \log \hat{\boldsymbol{y}}^T + (1-\boldsymbol{y}) \log\left(1-\hat{\boldsymbol{y}}\right)\right] \\ &= -\left[\boldsymbol{y} \log h_\theta(\boldsymbol{y}) + (1-\boldsymbol{y}) \log\left(1-h_\theta(\boldsymbol{y})\right)\right] \end{aligned} \tag{23}$$

where $h_\theta(\cdot)$ indicates the description of the whole network. The cross-entropy loss can improve the efficiency of the training process and is a suitable loss function for the unmixing task.

Moreover, in order to fit sparse unmixing, a row sparse $l_{2,1}$ regularization constraint on abundance maps $\boldsymbol{X}$ is introduced to the loss function. For the $i$th patch $\Omega_i$, the loss is calculated as

$$L_i = \lambda_1 \frac{1}{|\Omega_i|} \sum_{\boldsymbol{y}_p \in \Omega_i} \left( L_{\text{ce}}(\boldsymbol{y}_p, \hat{\boldsymbol{y}}_p) + \lambda_2 \left\| \hat{\boldsymbol{X}}_i^{\text{reshape}} \right\|_{2,1} \right). \tag{24}$$

The overall loss for $N$ patches in training sets is

$$L = \frac{1}{N} \sum_{i=1}^{N} L_i. \tag{25}$$

## IV. Experiments and Analysis

To evaluate the performance of the proposed method, two synthetic and two real HSI datasets are adopted in the experiment. Five traditional iterative sparse unmixing algorithms and two deep learning-based networks are chosen in this section. For traditional iterative methods, we choose SMP, SUnSAL, and its variant methods, namely, SUnSAL-TV and $S^2$WSU, to compare with the proposed method, which intuitively shows that the network derived by SUnSAL has superior performance than model-based methods. As for deep learning-based networks, SUnCNN and U-ADMM-BUNet are selected, the former being the only network proposed for sparse unmixing and the latter being the network used for semisupervised unmixing in [47]. It should be noted that U-ADMM-BUNet is not applied to sparse unmixing in [47], but due to its semisupervised training property and the fact that it derives from algorithm unrolling of SUnSAL, fixing its weights of the decoder to the values of the spectral library allows it to be used for sparse unmixing.

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### A. Evaluation Metrics

In this article, the analysis of the unmixing results is mainly based on quantitative analysis, and the merits and drawbacks of each method are reasonably compared through different objective evaluation metrics and then supplemented by subjective evaluation. For synthetic datasets, since the true abundance is known, the error between the estimated abundance and the true abundance can be measured by the signal-to-reconstruction error (SRE), the abundance root-mean-square error (aRMSE), and the abundance spectral angle distance (SAD), while for some of the real HSIs, due to the absence of true abundance, only the sparsity of the estimated abundance and the reconstruction root-mean-square error (rRMSE) between the reconstructed image and the original HSI is considered. Their specific expressions are given as follows:

$$\mathrm{SRE} = E\left[\|\boldsymbol{X}\|_2^2\right] / E\left[\|\boldsymbol{X} - \hat{\boldsymbol{X}}\|_2^2\right] \tag{26}$$

$$\mathrm{aRMSE} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left\|\hat{\boldsymbol{x}}_j - \boldsymbol{x}_j\right\|_2^2} \tag{27}$$

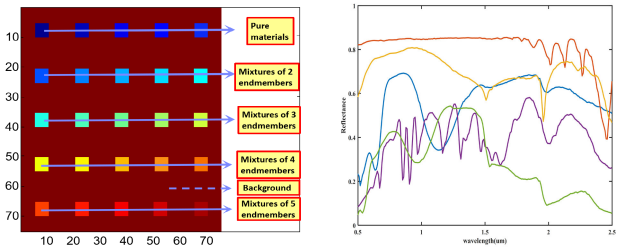
$$\mathrm{aSAD} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\boldsymbol{x}_i{}^T\hat{\boldsymbol{x}}_i}{\|\boldsymbol{x}_i\| \cdot \left\|\hat{\boldsymbol{x}}_i\right\|}\right) \tag{28}$$

$$\mathrm{rRMSE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{1}{L}\sum_{l=1}^{L}\left(\boldsymbol{y}_{i,l} - \hat{\boldsymbol{y}}_{i,l}\right)^2} \tag{29}$$

where $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$ denote real abundance and estimated abundance, respectively, and $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$ represent the input mixed pixel and reconstructed pixel, respectively. $N$ is the number of mixed pixels in HSIs and $L$ is the number of spectral bands. SRE is usually measured in decibels, that is, SRE (dB) $= 10\log_{10}\mathrm{SRE}$. From the definition, the larger the SRE value is, the closer the estimated abundance is to the true abundance, and the higher the unmixing accuracy of the algorithm. Sparsity is defined as the average number of nonzero values on each pixel, and in the actual calculation, a value greater than 0.001 is defined as a nonzero value. A smaller Sparsity value indicates that the actual number of endmembers is fewer and the abundance is sparser. When evaluating the unmixing results of real HSIs, it needs to be considered in combination with other evaluation indicators.

### B. Parameter Setting

The specific parameter settings of the proposed network are shown in Table I, and the network parameters are optimized using the Adam optimizer with the learning rate set to $1e^{-4}$. The setting of $\lambda$ in the network is consistent with the SUnSAL algorithm, and the values of $\lambda_1$ and $\lambda_2$ are, respectively, set to $5e^{-2}$ and $1e^{-4}$. In order to speed up the training of the network, the mini-batch strategy is adopted, and the image is divided into subimages of size $20 \times 20$ with overlapping areas and the batch size is set to 2. The optimization process of network parameters can also be regarded as an iterative procedure, and when the iteration is completed, the whole image is input to obtain the final abundance results. For other methods, their hyperparameters are adjusted to obtain the best unmixing performance in order to make the comparison of results more objective.




(a) (b)

Fig. 7. SD1. (a) Simulated image. (b) Selected endmembers.

### C. Experiments on Synthetic Datasets

In this section, the spectral library $\boldsymbol{A}$ we used in each method is generated from the USGS library [50], which contains 498 materials and 224 reflection values, ranging from 0.4 to 2.5 $\mu$m in the wavelength.

*1) Synthetic Dataset 1:* The Synthetic Dataset 1 (SD1) consists of five randomly selected endmembers from $\boldsymbol{A}$, following the LMM model, with a spatial size of $75 \times 75$ and 224 spectral bands. The abundance generation process is referred to the literature [51] and satisfies the ANC and ASC constraints, as shown in Fig. 7(a), where the background is a mixture of five endmembers, each with abundances of 0.1149, 0.0741, 0.2003, 0.2055, and 0.4051. The boxes in the first row are pure pixels, and the remaining 2–5 rows are the mixtures of 2–5 endmembers, respectively. Fig. 7(b) shows the spectral signatures of the five endmembers. After SD1 is generated, zero-mean white Gaussian noise is added with SNRs of 10, 20, 30, 40, and 50 dB.

The hyperparameters of all compared methods are tuned to the optimal, and Table II shows the specific SRE values for each algorithm. The unmixing performance of the proposed network outperforms the other compared algorithms at all levels of noise. Since the SUnSAL-TV and S$^2$WSU algorithms exploit the spatial information of the hyperspectral data, they both have higher unmixing accuracy than the SUnSAL algorithm, which only considers the abundance sparsity of individual pixels. It illustrates that it is necessary to make full use of the spatial information contained in the hyperspectral images in the unmixing task. In the SD1 experiments, the SRE results of the greedy algorithm SMP are comparable to the SUnSAL-TV algorithm, the U-ADMM-BUNet achieves suboptimal unmixing results, and the proposed DIFCNN network obtains the optimal unmixing performance.

The aRMSE, rRMSE, and aSAD results for SD1 are shown in Fig. 8, where aRMSE shows a trend consistent with the SRE results.

Combined with the detailed data, the SRE results of S$^2$WSU are 3–6 dB lower than SUnSAL-TV at SNR $\leq$ 30 dB and then 2–3 dB higher than SUnSAL-TV when SNR $\geq$ 40 dB. This is due to the fact that when there is more noise, the noise points affect the iteration of the spectral–spatial weight matrix in S$^2$WSU, making its calculation with larger errors, thus

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TABLE II
SRE (dB) RESULTS FOR DIFFERENT ALGORITHMS IN SD1

| SNR(dB) | SUnSAL | SUnSAL-TV | $S^2$WSU | SMP | SUnCNN | U-ADMM-BUNet | DIFCNN |
|---|---|---|---|---|---|---|---|
| 10 | 0.4105 | 3.6794 | 0.5756 | 4.1658 | 2.1214 | 6.4320 | **15.8984** |
| 20 | 3.7934 | 11.8764 | 5.2971 | 11.4165 | 11.0657 | 18.7311 | **22.9644** |
| 30 | 11.1513 | 21.8080 | 18.5130 | 21.0510 | 24.6376 | 28.6463 | **32.1945** |
| 40 | 20.7678 | 31.7377 | 33.7118 | 30.9168 | 30.8693 | 36.9055 | **41.1873** |
| 50 | 30.0992 | 40.4729 | 43.7491 | 40.0426 | 39.9632 | 48.6336 | **54.2263** |

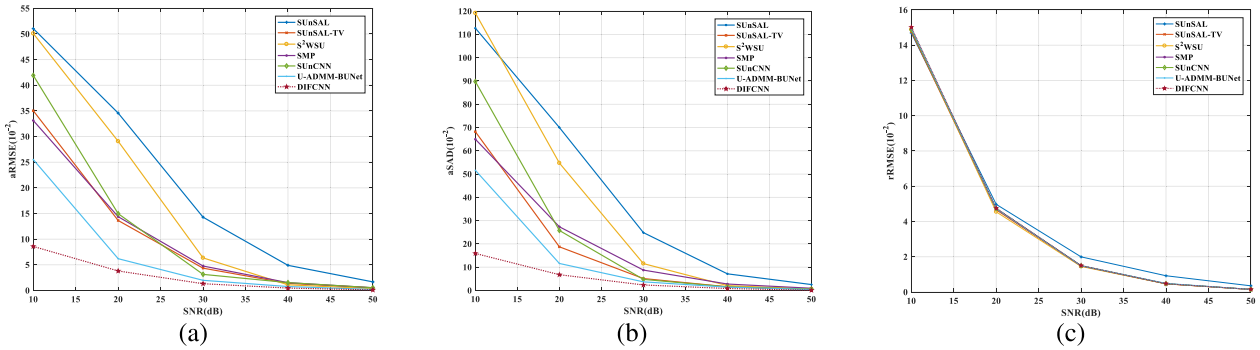



Fig. 8. Comparison of unmixing performance of each algorithm under different Gaussian noises (SD 1). (a) aRMSE. (b) aSAD. (c) rRMSE.

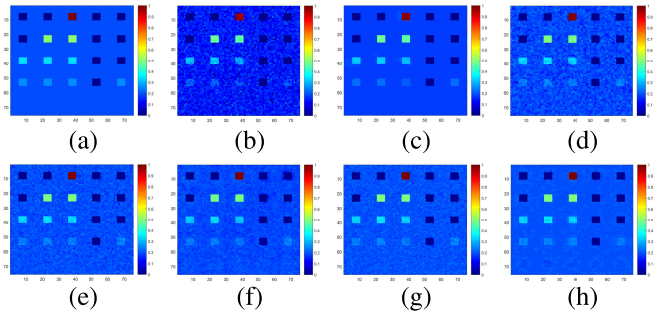



Fig. 9. Comparison of the 3rd endmember abundance maps at 30-dB Gaussian noise (SD1). (a) True abundance. (b) SUnSAL. (c) SUnSAL-TV. (d) $S^2$WSU. (e) SMP. (f) SUnCNN. (g) U-ADMM-BUNet. (h) DIFCNN.

affecting the unmixing performance. The U-ADMM-BUNet is an unfolding network for SUnSAL, and its SRE results are about 14.6 dB higher than SUnSAL on average and about 6.1 dB higher than the pure CNN SUnCNN, showing the great potential of using a network that combines model-driven with data-driven. In this article, we use CNN to realize the unfolding of the SUnSAL algorithm and consider the spatial information of HSIs so that the average SRE result is still about 5.4 dB higher than that of the U-ADMM-BUNet. From the experimental results, it can be seen that adding spatial correlation constraints to the SUnSAL algorithm, the TV regularization performs better when the SNR is low, and the spectral–spatial weight matrix outperforms when the SNR is high. With the benefit of noise reduction of CNN, the unmixing results of DIFCNN are more robust to noise.

Fig. 9 shows the abundance maps of the third endmember obtained by each algorithm under 30-dB Gaussian noise. Among them, the abundance map of SUnSAL algorithm contains more noise points, and when the pixels are highly mixed, it cannot distinguish each endmember properly. Although the SUnSAL-TV abundance map has the least noise points, some of the edges and regions with large abundance variations appear large errors because of their over-smooth abundance. The abundance map of $S^2$WSU, SMP, SUnCNN, and U-ADMM-BUNet is of little difference from the visual effect and has fewer noise points than the SUnSAL algorithm. The proposed DIFCNN has apparently fewer noise points in the background, indicating that the utilization of spatial information can reduce the noise points of the abundance results while preserving the edge contours and feature information of the abundance map.

*2) Synthetic Dataset 2:* The spatial size of Synthetic Dataset 2 (SD2) is 100 × 100 and also contains 224 spectral bands. The generation of SD2 follows the literature [51], where each pixel is a linear mixture of nine endmembers randomly selected from the spectral library $\boldsymbol{A}$. The spectra of nine endmembers are shown in Fig. 10(a), and Fig. 10(b)–(j) shows the corresponding true abundance maps for each endmember. The abundance coefficients of SD2 also satisfy the ANC and ASC constraints, vary smoothly in a spatial manner, and have edge characteristics that better represent the real ground. Five levels of zero-mean Gaussian white noise, i.e., 5, 10, 20, 30, and 40 dB, were added to the dataset.

The specified SRE values of each algorithm under different Gaussian noise are given in Table III. The proposed DIFCNN achieves the optimal SRE results in all Gaussian noise cases, reflecting the strong robustness to noise. When the SNR is 5 dB, the dataset is greatly affected by noise at this time, and the SRE results of all traditional algorithms are lower than 5 dB, it can be known that a large amount of noise will introduce errors to the established prior model, thus influencing the unmixing performance. $S^2$WSU has poor SRE results at low SNR, while the performance improves dramatically at high SNR. It can be observed that the spectral–spatial weighting model is more in line with the characteristics of HIS in the noiseless case but is heavily impacted by noise, and massive noise can seriously affect the unmixing accuracy.

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TABLE III
SRE (dB) RESULTS FOR DIFFERENT ALGORITHMS IN SD2

| SNR(dB) | SUnSAL | SUnSAL-TV | $S^2$WSU | SMP | SUnCNN | U-ADMM-BUNet | DIFCNN |
|---|---|---|---|---|---|---|---|
| 5 | 1.3231 | 3.5870 | 0.7073 | 1.6731 | 5.1484 | 3.6365 | **6.5760** |
| 10 | 3.5084 | 6.2836 | 1.7024 | 4.5189 | 11.6976 | 6.1484 | **13.9120** |
| 20 | 8.8121 | 13.0122 | 9.7935 | 17.2161 | 19.0192 | 17.0890 | **21.5554** |
| 30 | 16.0747 | 20.8656 | 22.1225 | 23.0200 | 26.0040 | 28.0457 | **29.5082** |
| 40 | 24.8639 | 29.7337 | 33.7698 | 31.9935 | 32.3066 | 34.4764 | **36.6686** |

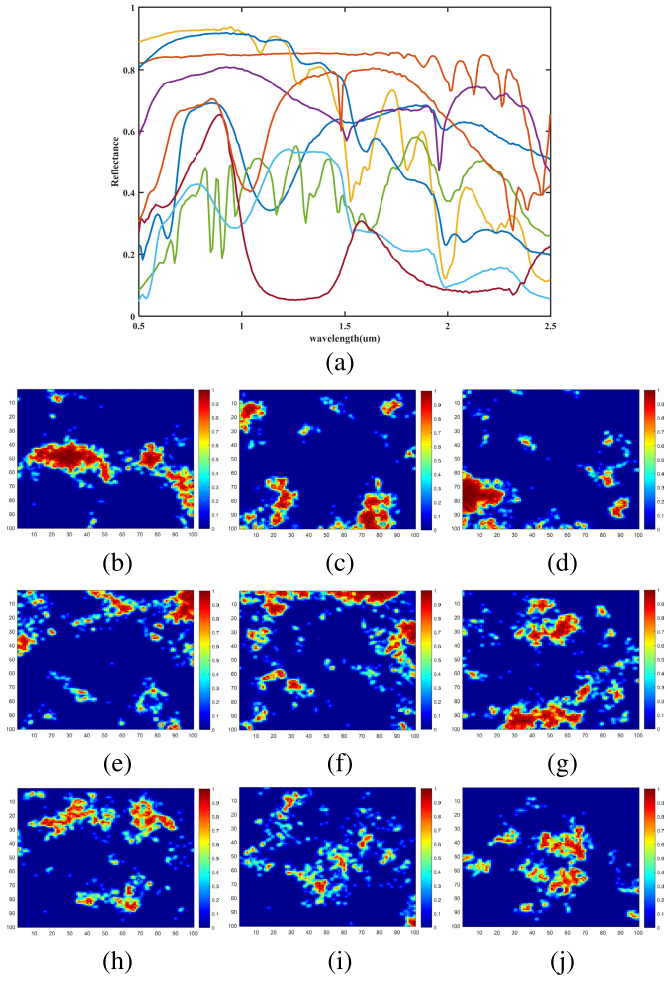



Fig. 10. SD2. (a) Nine endmembers spectra. (b) EM1. (c) EM2. (d) EM3. (e) EM4. (f) EM5. (g) EM6. (h) EM7. (i) EM8. (j) EM9.

The SRE values of the three deep learning-based networks are generally higher than those of the conventional algorithms, demonstrating the significant potential of deep learning in the field of unmixing. Compared with U-ADMM-BUNet, DIFCNN adds the learning of spatial information of HSI in the network, so the SRE is about 3.76 dB higher on average; compared with the CNN, DIFCNN introduces multiple shortcuts and skipping connections by unfolding the traditional algorithm, and the SRE results are about 2.80 dB higher than SUnCNN on average.

Fig. 11 shows the results of aRMSE, rRMSE, and aSAD for each algorithm in SD2 at five Gaussian noises. The aRMSE and aSAD results present performance consistent with the SRE results. At low SNR, the abundance estimated by the $S^2$WSU algorithm is less accurate than SUnSAL in these evaluation metrics. U-ADMM-BUNet achieves suboptimal performance under high SNR conditions, but the performance degrades faster when the noise interference is large, in contrast to the SUnCNN, which improves more steadily as the noise interference decreases. DIFCNN outperforms other algorithms in terms of aRMSE and rRMSE metrics, illustrating the advantages of the combined model- and data-driven network.

For the abundance results of each algorithm under 20-dB Gaussian noise, four endmembers are selected, and their absolute difference maps with the true abundance are shown in Fig. 12. In true abundance maps, a region with nonzero abundance value is considered as target and a region with zero abundance value is considered as background. The absolute differences of the SUnSAL, SUnSAL-TV, and $S^2$WSU algorithms are concentrated on the target region with a large error, while the differences of SMP, SUnCNN, U-ADMM-BUNet, and DIFCNN are mostly scattered in the background region with a smaller error. It indicates that the first three algorithms are affected by noise and are prone to map the abundance results to irrelevant endmembers, resulting in large errors in the target region. The SMP algorithm does not present the above error because the relevant endmembers are extracted first and most irrelevant endmembers are eliminated. The deep learning approaches can still obtain the same outcomes without removing endmembers in advance due to the powerful fitting ability of neural networks. SUnCNN is a 2-D CNN network that learns the global spatial information of HSI, and its learning of spatial information makes the error smoother overall, but there will be consistent errors in some local regions, which impacts the results of abundance estimation. In the U-ADMM-BUNet network; since it is a spectral-based unmixing method, only the spectral information of individual pixel is considered and the spatial information of the image is ignored, so the result has more noise points. DIFCNN utilizes spatial correlation in hyperspectral data on this basis to provide suppression of errors in the target, as well as suppression and smoothing of subtle errors in the background.

### D. Experiments on Real Datasets

*1) Jasper Ridge Dataset:* The Jasper Ridge dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), which initially contained 512 × 614 pixels. Each pixel records 224 reflectance values in the spectral band ranging from 380 to 2500 nm. As shown in Fig. 13(a), only a subimage with 100 × 100 pixels is considered in this section. After removing the bands impacted by dense water vapor and atmospheric effects, the image retains 198 spectral bands, and

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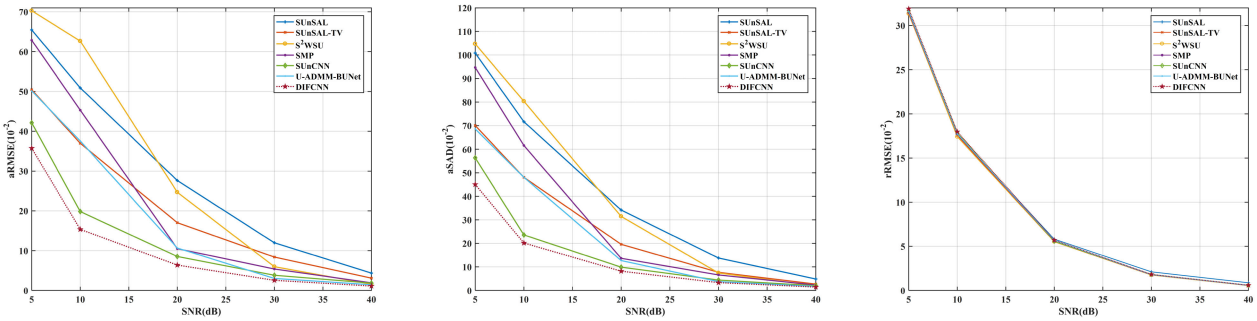



Fig. 11. Quantitative results of each algorithm under different Gaussian noises (SD2). (a) aRMSE. (b) aSAD. (c) rRMSE.

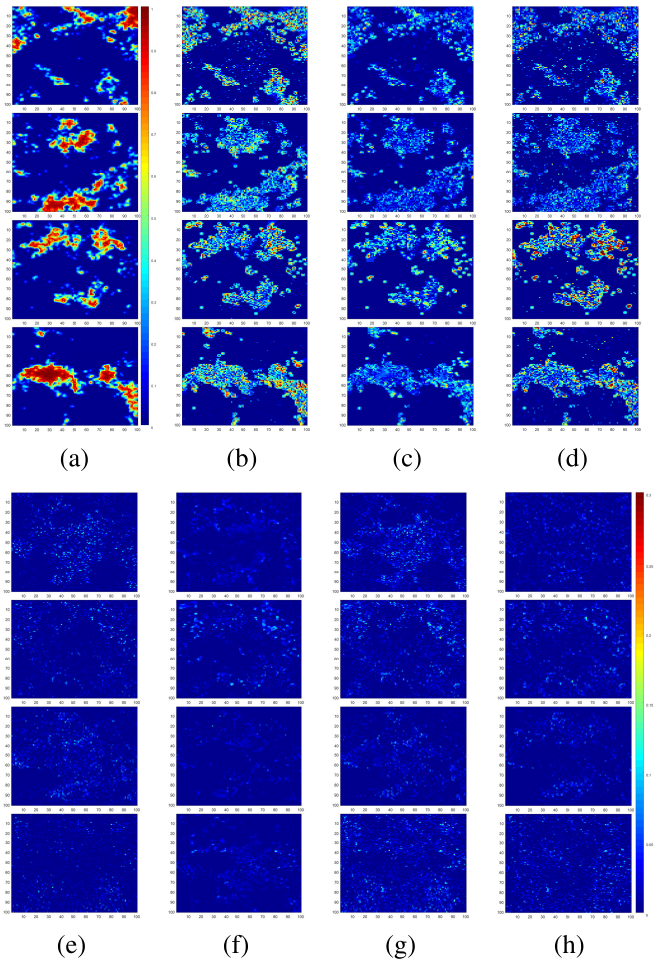



Fig. 12. True abundance and the absolute difference maps between the true abundance and the estimated abundance of SD2 (SNR = 20 dB). (a) True abundance. (b) SUnSAL. (c) SUnSAL-TV. (d) $S^2$WSU. (e) SMP. (f) SUnCNN. (g) U-ADMM-BUNet. (h) DIFCNN.

Fig. 13(b) shows four endmembers spectra manually selected from Jasper Ridge for Tree, Water, Dirt, and Road.

Since there is no matching spectral library for the Jasper Ridge dataset, we build a new spectral library for the Jasper Ridge dataset. The new spectral library contains 502 endmembers, four of which are the real spectral signals manually selected from the scene shown in Fig. 13(b), and the remaining 498 endmembers are all from spectral library $\boldsymbol{A}$ with the

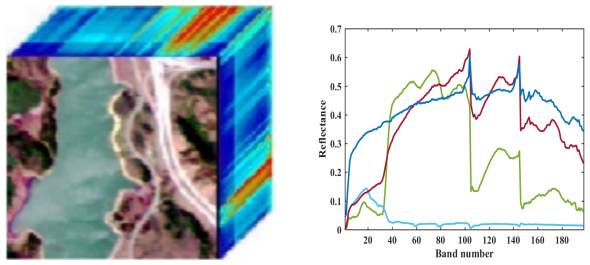



Fig. 13. Jasper Ridge dataset. (a) Image. (b) Endmembers are manually selected from the dataset.

removal of the contaminated bands, where the size of the spectral library is 198 × 502. The ability to distinguish true endmembers by the algorithm can be investigated by introducing a large number of irrelevant endmembers from the USGS spectral library into the new spectral library.

Table IV shows the SRE, aRMSE, aSAD, rRMSE, and Sparsity results of different algorithms on the Jasper Ridge dataset. Among them, the SRE, aRMSE, aSAD, and Sparsity of DIFCNN are, respectively, 10.9235, 0.5345, 0.2973, and 3.5041, which are better than other algorithms, displaying the effectiveness of exploiting model-driven concepts and spatial information in real HSI unmixing tasks. The aRMSE of SUnCNN is higher than that of $S^2$WSU, but $S^2$WSU obtains the smallest rRMSE value and the largest Sparsity, indicating that it introduces many irrelevant endmembers in reducing the error between the reconstructed image and the original HSI. Since the LMM does not accurately reflect the true distribution of materials well in a realistic scenario, which is unavoidable in sparse unmixing, better reconstruction does not mean better unmixing. Instead, DIFCNN has the highest rRMSE with the best SRE, which implies a poor reconstruction of the original hyperspectral image. At the same time, it has the lowest Sparsity, which means that the obtained abundance matrix is the sparsest, that is, the hyperspectral image can be reconstructed using fewer endmembers.

The ground-truth and estimated abundance maps of four endmembers are shown in Fig. 14. Combined with the quantitative analysis results in Table IV, we can see that although the abundance of U-ADMM-BUNet is sparse, it cannot map the abundance of the "Road" to the correct endmember and therefore has a great impact on the unmixing results. For the

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TABLE IV
SRE, aRMSE, aSAD, rRMSE, AND SPARSITY RESULTS FOR JASPER RIDGE DATASET. THE OPTIMAL RESULTS ARE IN BOLD

| | SUnSAL | SUnSAL-TV | $S^2$WSU | SMP | SUnCNN | U-ADMM-BUNet | DIFCNN |
|---|---|---|---|---|---|---|---|
| SRE | 6.7164 | 8.7361 | 8.2427 | 7.6235 | 7.3092 | 7.2064 | **10.9235** |
| aRMSE | 0.6555 | 0.5458 | 0.5474 | 0.5830 | 0.5832 | 0.5959 | **0.5345** |
| aSAD | 0.4350 | 0.3916 | 0.4194 | 0.4319 | 0.4561 | 0.4827 | **0.2973** |
| rRMSE | 0.0158 | 0.0089 | **0.0067** | 0.0151 | 0.0161 | 0.0174 | 0.0249 |
| *Sparsity* | 7.9960 | 5.1024 | 14.2662 | 4.2833 | 8.6780 | 3.6511 | **3.5041** |

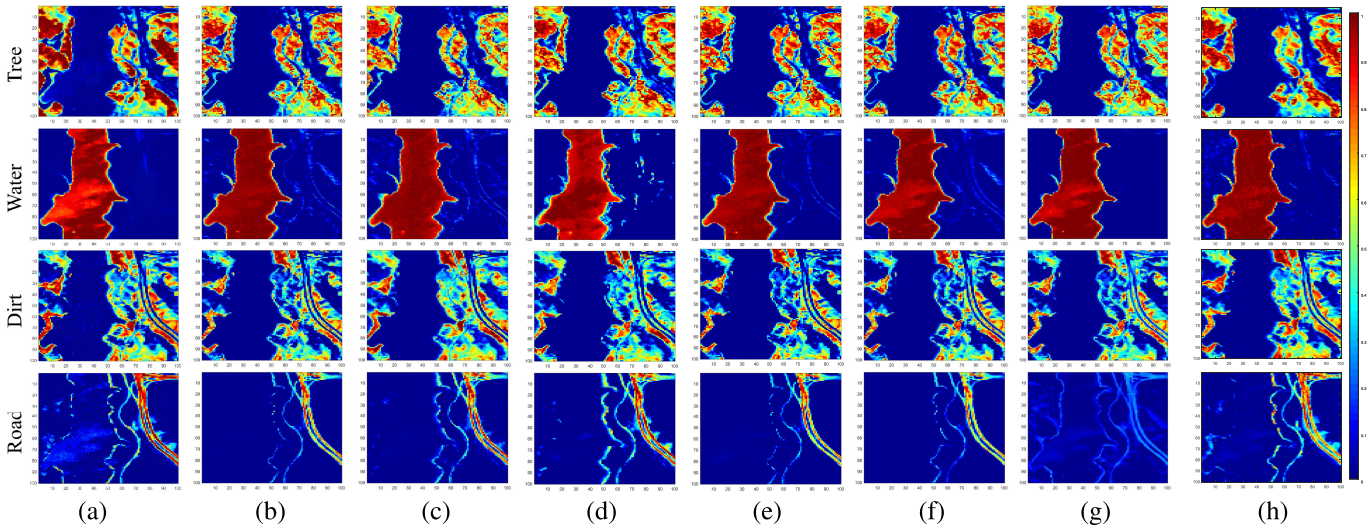



Fig. 14. Ground-truth and abundance maps estimated by different methods for the Jasper Ridge dataset. (a) GT. (b) SUnSAL. (c) SUnSAL-TV. (d) $S^2$WSU. (e) SMP. (f) SUnCNN. (g) U-ADMM-BUNet. (h) DIFCNN.

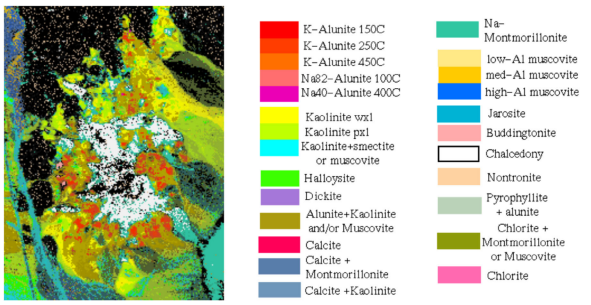



Fig. 15. Distribution of different minerals in the Cuprite dataset.

abundance maps of "Dirt," SUnSAL-TV fits better with the ground truth. SUnCNN has smoother abundance maps due to the 2-D CNN. None of the above methods can distinguish well between "Water" and "Road" because the spectral curves of these two endmembers are very similar. In addition, there may be irrelevant endmembers in the spectral library that are similar to the true endmembers' spectra, which may bring interference to the abundance results. The "Water" abundance map of DIFCNN has more noise points, while the abundance maps of "Tree," "Road," and "Soil" have the highest similarity to the reference abundance and contain more details and edge information than other algorithms.

*2) Cuprite Dataset:* The AVRIS Cuprite dataset covers the mining area in Las Vegas, NV, USA, and is a common benchmark dataset for evaluating the unmixing performance. As shown in Fig. 15, a submap with a size of 250 × 191 is selected for the experiment. To avoid the influence on the accuracy of the unmixing results, the noisy channels and water absorption channels are removed, and 188 spectral bands were retained. There are 14 minerals in this region, and minor differences between variants of similar minerals eventually reduced the number of endmembers to 12, which are Alunite, Andradite, Buddingtonite, Dumortierite, Kaolinite1, Kaolinite2, Muscovite, Montmorillonite, Nontronite, Pyrope, Sphene, and Chalcedony. As the endmembers in the Cuprite dataset are contained in the spectral library given by USGS, we directly use the library $\boldsymbol{A}$ with 498 endmembers. In the experiments, polluted bands of $\boldsymbol{A}$ are removed and a spectral library containing 188 bands was obtained. The Cuprite dataset lacks ground-truth abundance as the reference, so the classification maps generated by Tricorder 3.3 are used as the reference for the qualitative analysis in this section. For the quantitative analysis, again due to the lack of true abundance information, the unmixing results for the Cuprite dataset are analyzed only in terms of the rRMSE and the Sparsity of the abundance.

Table V shows the rRMSE and Sparsity results of different algorithms on the Cuprite dataset. The hyperparameters of each algorithm are tuned to the optimum in the experiments. The Sparsity obtained by each method is larger in the face of a more complex realistic scenario. In addition, the presence of numerous endmembers with high similarity in the spectral library further increases the difficulty of achieving lower Sparsity. The SMP algorithm has the lowest Sparsity because it is a greedy algorithm that already eliminates many invalid endmembers beforehand. SUnSAL-TV and $S^2$WSU have much better rRMSE results than SUnSAL, indicating that the reconstruction can be improved by utilizing the spatial information

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TABLE V
QUANTITATIVE UNMIXING RESULTS FOR CUPRITE DATASET. THE OPTIMAL RESULTS ARE IN BOLD

| | SUnSAL | SUnSAL-TV | $S^2$WSU | SMP | SUnCNN | U-ADMM-BUNet | DIFCNN |
|---|---|---|---|---|---|---|---|
| rRMSE | 0.0169 | 0.0054 | **0.0053** | 0.0061 | 0.0059 | 0.0071 | 0.0067 |
| *Sparsity* | 18.5319 | 20.6533 | 25.9599 | **15.5502** | 27.3406 | 18.9873 | 18.1365 |

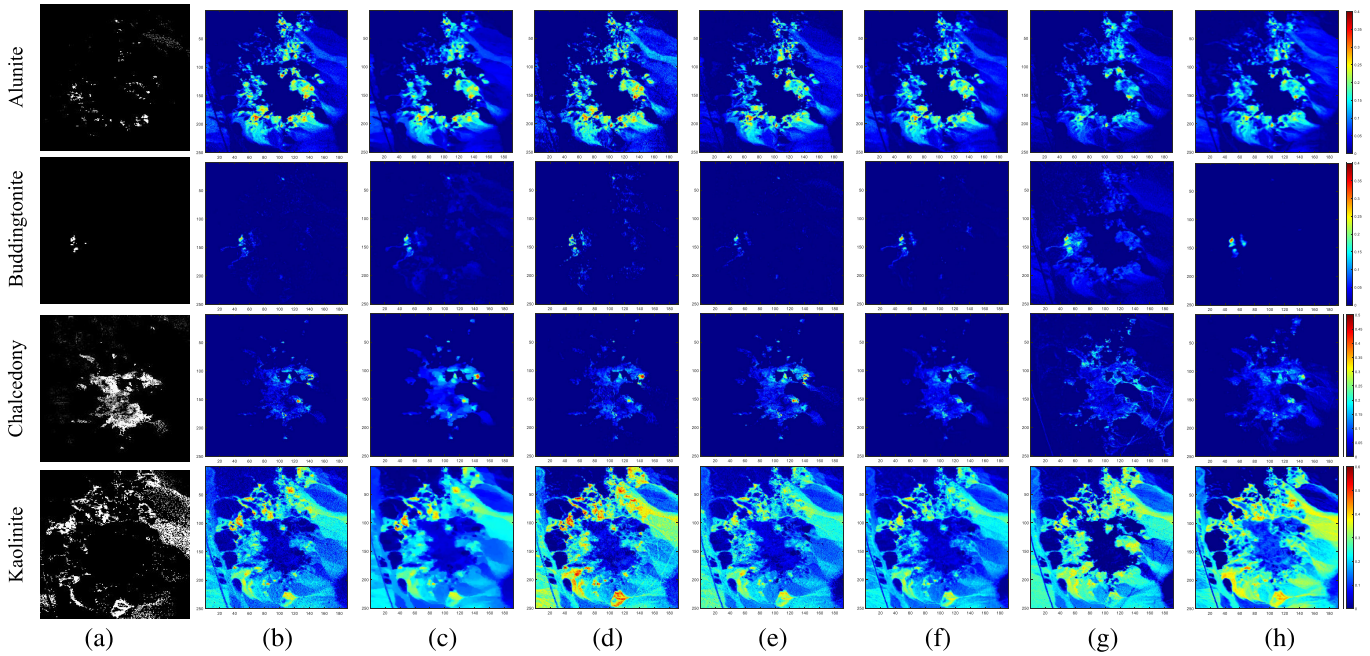



Fig. 16. Abundance maps of four materials estimated by different methods for the Cuprite dataset. (a) Tricorder 3.3. (b) SUnSAL. (c) SUnSAL-TV. (d) $S^2$WSU. (e) SMP. (f) SUnCNN. (g) U-ADMM-BUNet. (h) DIFCNN.

of HSI. However, the extraction of spatial information introduces a rise in Sparsity because the priori information on neighborhood similarity leads to the inclusion of more endmembers in each pixel. SUnCNN has the largest Sparsity due to the lack of sparsity constraint on the abundance, and in other words, the estimated abundance contains too much interference from irrelevant endmembers. U-ADMM-BUNet is motivated by the traditional sparse algorithm model and obtains a sparser result. The proposed DIFCNN has an rRMSE of 0.0067 and a Sparsity of 18.1365, obtaining a smaller sparsity while minimizing the reconstruction error.

Select four endmembers of these abundance maps for visual comparison, as shown in Fig. 16, and the first column is the mineral distribution map generated by Tricorder3.3 software. Compared with SUnSAL, the abundance results obtained by SUnSAL-TV are smoother and have fewer noise points, but at the same time, the edge contour and detailed information are absent. The abundance results of $S^2$WSU and SMP have more noise but are clearer at the edge contours. The abundance maps of SUnCNN have less noise, reflecting the role of the convolutional layers in smoothing the abundance results. The abundance maps of Buddingtonite and Chalcedony from U-ADMM-BUNet differ significantly from the reference and also contain abundance information of other similar endmembers, reflecting a poor ability to identify valid endmembers from the spectral library. This is due to the fact that the network is unfolded by a traditional sparse algorithm, and the sparsity constraint contained in the structure concentrates the abundance results on fewer endmembers, thus merging the abundance results of some similar endmembers and eventually leading to a certain error. For the same reason, the same error also occurs on the abundance map of Kaolinite obtained by DIFCNN. However, other abundance obtained by DIFCNN is more consistent with the distribution of real ground features, and the abundance map of Buddingtonite contains the fewest noise points and is closest to the reference. While there are fewer noise points, DIFCNN retains edge information and detailed features. This demonstrates that incorporating convolutional layers into the algorithm unrolling network has two advantages. First, it reduces noise in the abundance results. Second, extracting spatial information helps address the issue of the spectral-based network U-ADMM-BUNet's inability to identify effective endmembers. However, the performance of DIFCNN is noticeably lower on real datasets compared to simulated datasets. We attribute this to the inherent limitations of sparse unmixing. Sparse unmixing is affected by the spectral variability and misfit of LMM in real-world environments due to its reliance on a fixed spectral library and LMM.

### E. Analysis

*1) Hyperparameter Analysis:* Based on the inference of the algorithm unrolling, the hyperparameter $\lambda$ determines the initialization of the learnable parameters in the network. We selected two simulated datasets with SNR of 20 dB for experiments, and Fig. 17 shows the results. From Fig. 17,

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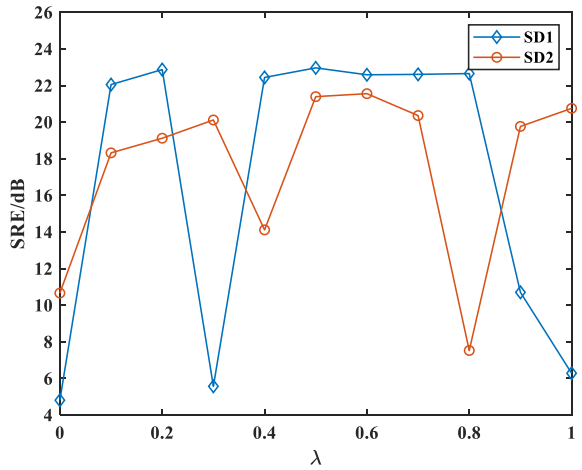



Fig. 17. Abundance estimation performance of different $\lambda$'s under the SNR of 20 dB (SD1 and SD2).

we can observe that the stability of the SRE results can be maintained when $\lambda$ is within a certain range, which is due to the adaptive adjustment of the parameters by the neural network, unlike the traditional algorithm where the parameters are fixed. When $\lambda$ is out of the range interval, the unmixing performance of the network degrades significantly, which validates the need for proper initialization of the network based on algorithm unrolling. However, the range interval is variable, which is attributed to the fact that different initialization can lead the network to different local optima. In addition, it should be noted that SD1 exhibits a wider SRE vibration range compared to SD2, which can be attributed to differences in data volume and the number of endmembers. With respect to the upper limit, SD2 has more endmembers and pixels than SD1, making it more challenging to achieve optimal outcomes in training. In terms of the lower limit, from a data-driven perspective, SD1 is more likely to yield inferior unmixing results due to improper initialization as the amount of training data is less than that of SD2.

*2) Layer Analysis:* Literature [47] has given the impact of the number of iterative layers on abundance estimation performance. The results show that for an unrolling-based unmixing network, the number of layers does not have a significant impact on the final performance. However, this is the conclusion obtained for iterative layers of the same construction. Since the multiscale strategy is applied in our network, that is, different convolutional kernel sizes are implemented in different layers, we compare networks with the number of layers from 1 to 4, where the convolutional kernel size in the $k$th layer from the bottom is $(2k-1) \times (2k-1)$. Table VI shows the unmixing performance of the network with different numbers of layers on the dataset with an SNR of 20 dB in SD2. Clearly, the multiscale strategy brings the benefit of fusing spatial multiscale information but limits the number of layers at the same time. For the network with four layers, a convolutional kernel of size $7 \times 7$ requires more padding to ensure invariance on the spatial scale, introducing much unnecessary spatial information. In addition, deeper network layers increase the difficulty of training.

TABLE VI

IMPACT OF NUMBER OF LAYERS ON ABUNDANCE ESTIMATION PERFORMANCE UNDER THE SNR OF 20 dB (SD2). THE OPTIMAL RESULTS ARE IN BOLD

| number of layers | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| SRE(dB) | 12.3582 | 19.3819 | **21.5554** | 15.3094 |

TABLE VII

IMPACT OF FEATURE EXTRACTION SCALE ON ABUNDANCE ESTIMATION PERFORMANCE UNDER THE SNR OF 20 dB (SD2). THE OPTIMAL RESULTS ARE IN BOLD

| scale | $1 \times 1$ | $3 \times 3$ | $5 \times 5$ | multi-scale |
|---|---|---|---|---|
| SRE(dB) | 17.8692 | 20.4073 | 21.3122 | **21.5554** |

TABLE VIII

PARAMETER SIZE AND RUNNING TIME OF DIFFERENT DEEP LEARNING-BASED METHODS UNDER THE SNR OF 20 dB (SD2). THE OPTIMAL RESULTS ARE IN BOLD

| | U-ADMM-BUNet | SUnCNN | DIFCNN |
|---|---|---|---|
| parameter size(MB) | **4.97** | 7.26 | 52.26 |
| running time(s) | **0.5244** | 1.8714 | 5.7712 |

*3) Feature Extraction Scale Analysis:* To showcase the effectiveness of the multiscale strategy, we conducted a comparison between DIFCNN and other networks that extract features at different scales: $1 \times 1$, $3 \times 3$, and $5 \times 5$. In this comparison, we replaced the kernels of the feature extraction convolutional layers in DIFCNN with convolutional kernels of corresponding sizes. The results of our experiments are shown in Table VII. Network with a scale of $1 \times 1$ performs the worst, primarily due to its limited ability to extract spatial features. As we increased the feature extraction scale, the unmixing performance of the network improved. However, this improvement came at the expense of an increase in the number of parameters. Although the network with a scale of $5 \times 5$ produces comparable results to the multiscale, it has far more parameters than the multiscale network. Benefiting from more comprehensive and detailed information being captured, DIFCNN can have excellent unmixing performance with a relatively small number of parameters.

*4) Parameter Size and Running Time Analysis:* As a result of using convolutional layers with larger size convolutional kernels, we face certain drawbacks in terms of parameter size and running time (excluding training time), as shown in Table VIII. However, due to the fast processing capabilities of modern GPUs, it is now feasible to prioritize a more intricate neural network in order to achieve improved unmixing outcomes.

## V. CONCLUSION

In this work, we have introduced an interpretable CNN structure for hyperspectral sparse unmixing that combines the advantages of both data- and model-driven methods. The proposed network unfolds the traditional iterative algorithm and incorporates convolutional layers to automatically learn spatial features, trained in an end-to-end manner. Moreover, nonlinear layers are imposed after the convolution operation, which speeds up the convergence of the network without affecting

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the interpretability of the network. Evaluation experiments on synthetic and real data demonstrate that our approach achieves better performance than both iterative sparse unmixing methods and learning-based sparse unmixing methods, especially at low signal-to-noise ratios, showing strong robustness to noise. We will focus on researching network structures that better combine model- and data-driven advantages in the future.

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